Lab 9 Logistic and Count-Variable Regression

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For the logistic regression:

```
Model Accuracy:  68.42%
              precision    recall  f1-score   support

           0       0.69      0.96      0.80        25
           1       0.67      0.15      0.25        13

    accuracy                           0.68        38
   macro avg       0.68      0.56      0.53        38
weighted avg       0.68      0.68      0.61        38
```

The model's accuracy is 68.42 percent. The lowest precision value is 0.67 in class 1. None of the categories have at least 30 support. I did not obtain a pseudo-R^2 value since the model is limited in capabilities.

For the count-variable regression:

1. Your model should include 4 significant predictors. What are they?
2. Which of these predictors is the most significant (hint: look at the 'coef' column)
3. Do any of your significant predictors have an overinflated VIF (VIF > 10)?
4. How much of the variability is captured by the model (hint: look at pseudo-r^2)

```
Skipped 844 row(s) missing the target or a predictor
             Feature       VIF
               const  1.000000
     binge_crudeprev  2.528842
    checkup_crudeprev  1.750444
   csmoking_crudeprev  6.934794
depression_crudeprev  2.216376
    obesity_crudeprev  2.573017
      sleep_crudeprev  3.209334
  selfcare_crudeprev 30.514413
 indeplive_crudeprev 30.953309
loneliness_crudeprev  3.432565
foodinsecu_crudeprev 62.830787
housinsecu_crudeprev 46.839954
----------------------------------------
Fitting model...
Optimization terminated successfully.
         Current function value: 2.347799
         Iterations 4
```

```
                          Poisson Regression Results
==============================================================================
Dep. Variable:        mhlth_crudeprev   No. Observations:                 2299
Model:                        Poisson   Df Residuals:                     2287
Method:                           MLE   Df Model:                           11
Date:                Sun, 25 Oct 2026   Pseudo R-squ.:                0.04356
Time:                        16:26:44   Log-Likelihood:                -5397.6
converged:                       True   LL-Null:                       -5643.4
Covariance Type:            nonrobust   LLR p-value:                1.908e-98
============================================================================================
                               coef    std err          z      P>|z|      [0.025      0.975]
--------------------------------------------------------------------------------------------
Intercept                    2.8313      0.005    557.495      0.000       2.821       2.841
binge_crudeprev              0.0077      0.008      0.954      0.340      -0.008       0.024
checkup_crudeprev           -0.0013      0.007     -0.192      0.848      -0.015       0.012
csmoking_crudeprev           0.0142      0.013      1.100      0.271      -0.011       0.040
depression_crudeprev         0.0436      0.008      5.800      0.000       0.029       0.058
obesity_crudeprev            0.0006      0.008      0.076      0.940      -0.015       0.017
sleep_crudeprev              0.0096      0.009      1.059      0.289      -0.008       0.027
selfcare_crudeprev          -0.0733      0.027     -2.708      0.007      -0.126      -0.020
indeplive_crudeprev          0.1032      0.027      3.759      0.000       0.049       0.157
loneliness_crudeprev         0.0345      0.009      3.717      0.000       0.016       0.053
foodinsecu_crudeprev        -0.0492      0.039     -1.247      0.212      -0.126       0.028
housinsecu_crudeprev         0.0613      0.034      1.786      0.074      -0.006       0.129
============================================================================================
```

There are 4 significant predictors: depression_crudeprev, selfcare_crudeprev, indeplive_crudeprev, and loneliness_crudeprev. The most significant predictor is indeplive_crudeprev which has a coefficient of 0.1032. Two of the significant predictors have VIF's greater than 10, and they are selfcare_crudeprev (30.51) and indeplive_crudeprev (30.95). The pseudo R^2 value tells how much variability is captured by the model, and about 4.36 percent is captured.